Smart Clinic AI Assistant

Project Info Form

DEPI — Digital Egypt Pioneers Initiative | Track: AI & Data Science

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Project name: Smart Clinic AI Assistant

Project Idea: An n8n-based automation system for a single-doctor clinic that handles patient communication end-to-end via a Telegram chatbot. All routing between general inquiries and appointment booking is done through Telegram inline-button menus (no LLM involved in that step), with real-time Google Calendar availability checks to prevent double-booking. The system also sends 24-hour interactive appointment reminders, generates an automated daily financial report (calculated in code, not by the LLM), and collects post-visit patient feedback, which an LLM classifies by sentiment — the only step in the system that uses an LLM. This reduces manual staff work and gives the doctor accurate, up-to-date visibility into clinic performance.

Github repo URL: https://github.com/nhahub/NHA-5-201

Tools, Libraries and APIs

- n8n (core automation/orchestration engine)
- Telegram Bot API
- Google Calendar API
- Google Sheets API
- Gmail API
- LLM API (feedback sentiment classification only)

Project Team Members

- Ahmed Hossam Abdullah
- Mohamed Alaa Ibrahim
- Ziad Ahmed Fathy
- Ahmed Mossa Anwar

Team Leader

- Ahmed Khaled Abdelfattah

Final Project Deliverable: A working n8n automation system (not a standalone app) — a set of deployed, interconnected n8n workflows exposed to patients through a Telegram bot, demoed live rather than packaged as an api/mobile/web app.

Project Pipeline:

1 Patient sends a message via Telegram.

2. Button menu routes the patient to Inquiry or Booking (no LLM).
3. Booking path checks Google Calendar for availability (code-based, no double-booking) before confirming — single doctor, 20-minute slots.
4. Confirmed data (patient, appointment, chat_id) is logged to Google Sheets — the shared data store.
5. From Sheets, three scheduled jobs run independently: Reminders (sent 24h before each appointment); Financial report (daily revenue and case count, calculated in code and emailed to the doctor); Feedback (LLM sentiment analysis on post-visit patient responses).

(Recreate this flow as a diagram in diagrams.net per the submission requirement — the stages above are the exact content to lay out.)

Dataset source: No fixed external dataset — the system operates on live data generated during operation (bookings, appointments, and feedback captured through Telegram/Calendar/Sheets), not pre-collected training data.

Timeline Deliverables

Week 3 & 4:

Weeks deliverables (project-wise):

- **System Analysis & Design package (due 6 Nov 2026):** problem statement and objectives; Use Case diagram and descriptions; software architecture (n8n workflows connected to Telegram, Google Calendar, Google Sheets, Gmail); ER diagram with logical and physical schema (Bookings, Daily Ledger, Feedback); DFD (context level and level 1); Sequence diagram for the booking flow; Activity and State diagrams (booking status: Pending, Confirmed, Cancelled, Needs Rescheduling); wireframes of the bot's conversation flows with UI/UX guidelines; technology stack, component diagram, and deployment diagram.
- **Booking bot stabilization:** fix for entering name and age without replying to the bot's message (conversation state kept per chat_id); bot token moved to n8n credentials and removed from the exported workflow; clear, descriptive node names.
- **Google Sheets integration:** every confirmed booking is logged (appointment_id, chat_id, name, age, date, time, status).
- **Module 2 started:** scheduled reminder workflow, 24 hours before each appointment, with Confirm / Reschedule / Cancel buttons.
- **Testing:** test cases for the booking flow and the first bug report.

Team Members' Roles:

- Ahmed Khaled Abdelfattah (Team Leader) role: coordinates the System Analysis & Design package (architecture, Use Case diagram, ER diagram review), organizes the GitHub repository, and follows up on deadlines
- Ahmed Hossam Abdullah role: Google Sheets integration for booking logs, and the Module 2 reminder workflow

- Mohamed Alaa Ibrahim role: stabilizes the Telegram booking bot (conversation state, token security, node naming) and starts the daily financial report workflow
- Ziad Ahmed Fathy role: test cases and bug reports for the booking flow, and the Sequence diagram
- Ahmed Mossa Anwar role: DFD, Activity and State diagrams, and conversation-flow wireframes with UI/UX guidelines